

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology	
Subject: Digital Signal and Image Processing(01CT1513)	Simulate smoothing and sharpening operation on images	
Experiment-9	Date:	Enrollment No:92400133138

Aim:- Simulate smoothing and sharpening operation on images using spatial filters.

Objective:

To reduce image noise using smoothing filters and enhance edges and fine details using sharpening filters.

Procedure:

1. Load the input image.
2. Apply a smoothing filter using a suitable kernel size.
3. Apply a sharpening filter using a sharpening kernel.
4. Display the original, smoothed, and sharpened images side by side.
5. Save the filtered images.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
def apply_smoothing_filter(image, kernel_size):
    smoothed_image = cv2.blur(image, (kernel_size, kernel_size))
    return smoothed_image
def apply_sharpening_filter(image):
    kernel = np.array([[0, -1, 0], [-1, 5, -1], [0, -1, 0]])
    sharpened_image = cv2.filter2D(image, -1, kernel)
    return sharpened_image
image_path = 'img_10.jpeg'
input_image = cv2.imread(image_path)
smoothed_image = apply_smoothing_filter(input_image, kernel_size=5)
sharpened_image = apply_sharpening_filter(input_image)
combined_image = np.hstack((input_image, smoothed_image, sharpened_image))
cv2.imshow("Original | Smoothed | Sharpened", combined_image)
cv2.waitKey(0)
smoothed_path = 'smoothed_image.jpg'
sharpened_path = 'sharpened_image.jpg'
cv2.imwrite(smoothed_path, smoothed_image)
cv2.imwrite(sharpened_path, sharpened_image)
```

```
print(f"Smoothed image saved at: {smoothed_path}")
print(f"Sharpened image saved at: {sharpened_path}")
```

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Output:



Advantages:

1. Smoothing reduces noise and unwanted fine details, while sharpening enhances edges and important image details.
2. Both operations improve the visual quality of images and are simple to implement using spatial filters.

Applications:

Used in image enhancement, noise reduction, medical imaging, computer vision, photography, object detection, and edge detection.

Conclusion:

Smoothing and sharpening operations were successfully performed using spatial filters.

The smoothing filter reduced noise, while the sharpening filter enhanced edges and fine details, demonstrating their usefulness in image enhancement.

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Github Link:

<https://github.com/ayush-subudhi/Dsip-Experiment>